

In[178]:=

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(* Frequency and target field *)
frequencyTTF = 200*^3; (* 200 kHz *)
targetPeakField = 200; (* V/m peak without rings *)
phaseTTF = 0;

(* Electrodes *)
electrodeLeft = {-0.35, 0};
electrodeRight = {0.35, 0};

(* Base TTF field (no scaling yet) *)
baseField[x_, y_, t_] :=
Cos[2 Pi*frequencyTTF*t + phaseTTF]*
(1/(Norm[{x, y} - electrodeLeft] + 0.01) -
1/(Norm[{x, y} - electrodeRight] + 0.01));

(* Find max field magnitude across torso (snapshot at t=0) *)
sampleProfile = Table[Abs[baseField[r, 0, 0]], {r, -0.3, 0.3, 0.01}];
peakValue = Max[sampleProfile];

(* Scale so max = 200 V/m *)
scaleFactor = targetPeakField/peakValue;
ttfField[x_, y_, t_] := scaleFactor*baseField[x, y, t];

(* Nested laminated iron rings *)
chainRadii = {0.2, 0.25, 0.3};
chainSegments = 20;
chainPositionsAll = Flatten[
Table[
Table[{r*Cos[ $\theta$ ], r*Sin[ $\theta$ ]}, { $\theta$ ,
Table[2 Pi i/chainSegments, {i, 0, chainSegments - 1}]}],
{r, chainRadii}], 1];

(* Scattering from laminated rings *)
scatteredField[x_, y_, t_] :=
Total[
Table[
-0.25*ttfField[pos[[1]], pos[[2]], t]/(Norm[{x, y} - pos] + 0.01),
{pos, chainPositionsAll}
]
];
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totalFieldNoRings[x_, y_, t_] := ttffield[x, y, t];
totalFieldWithRings[x_, y_, t_] := ttffield[x, y, t] + scatteredField[x, y, t];

(* Compare center field over time *)
timePoints = Table[t, {t, 0, 5*^-6, 5*^-8}];
centerNoRings = Table[{t, totalFieldNoRings[0, 0, t]}, {t, timePoints}];
centerWithRings = Table[{t, totalFieldWithRings[0, 0, t]}, {t, timePoints}];

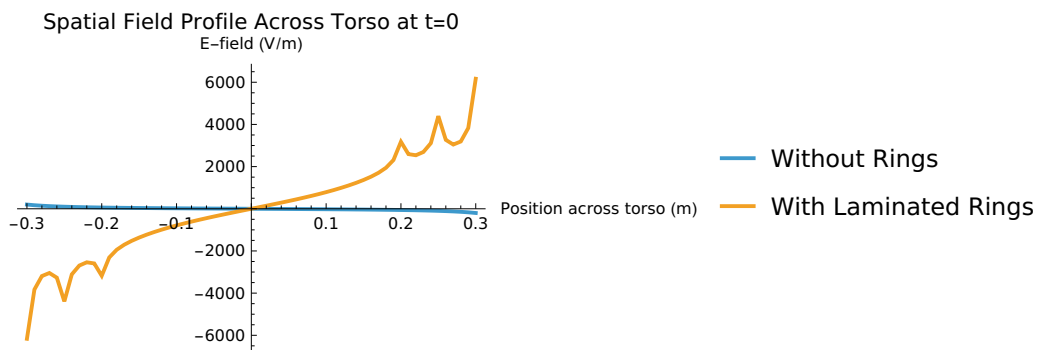
(* Compare field across torso at t = 0 *)
profileNoRings = Table[{r, totalFieldNoRings[r, 0, 0]}, {r, -0.3, 0.3, 0.01}];
profileWithRings = Table[{r, totalFieldWithRings[r, 0, 0]}, {r, -0.3, 0.3, 0.01}];

ListLinePlot[{profileNoRings, profileWithRings},
  PlotRange → All,
  PlotLegends → {"Without Rings", "With Laminated Rings"},
  AxesLabel → {"Position across torso (m)", "E-field (V/m)"},
  PlotLabel → "Spatial Field Profile Across Torso at t=0"]

ListLinePlot[{centerNoRings, centerWithRings},
  PlotRange → All,
  PlotLegends → {"Without Rings", "With Laminated Rings"},
  AxesLabel → {"Time (s)", "Field at Center (V/m)"},
  PlotLabel → "Center Field Comparison: No Rings vs Laminated Russian Doll"]

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Out[199]=



Out[200]=

Center Field Comparison: No Rings vs Laminated Russian Doll
Field at Center (V/m)

